

RelGate: Evidence-Grounded Readiness Gates for LLM-Reviewed Cloud Changes

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Abstract—Cloud-change review asks whether a release has sufficient evidence for observability, alerting, rollout, rollback, ownership, service impact, and validation. LLMs can assist with this review, but unstructured responses may block well-documented changes or claim readiness without support. RelGate reviews a fixed change bundle through seven configurable gates. Every PASS must quote a verbatim span from the bundle. A gate without such support is marked `MISSING_EVIDENCE`. We evaluated 12 synthetic scenarios (9 unsafe and 3 `READY` controls) across three models and three review modes (freeform, checklist, and RelGate), with one temperature-0 call per case/model/mode (108 calls). After strict rescoreing of the frozen outputs, none of the modes approved any of the 27 unsafe observations per mode. Freeform blocked 9/9 `READY` controls, checklist blocked 1/9, and RelGate blocked 0/9. RelGate’s mean per-response unsupported-claim fraction was 0.056, compared with 0.208 for checklist-only review. Its observed decision accuracy was 1.000. In this small controlled study, RelGate made passing judgments traceable to the bundle and did not show the false blocks seen in the other modes. Larger studies with realistic change records are needed before drawing conclusions about production use.

Index Terms—software reliability, LLM-assisted review, evidence grounding, cloud changes

I. MOTIVATION

Reliable release processes require explicit information about monitoring, rollout, rollback, ownership, and validation [1], [2]. One fully specified `READY` control included dashboard and alert links, a staged rollout, rollback triggers, ownership, SLO analysis, and test results. Even so, all three models blocked this control under freeform review. Checklist reviews also returned PASS judgments with no supporting text in the bundle. We distinguish this failure from the broader problem of LLM hallucination [3]. RelGate treats this as evidence auditing rather than open-ended advice.

RAG adds retrieved, non-parametric context to a prompt [4]. Chain-of-Verification asks the model to check an earlier answer through verification questions [5]. Structured-generation methods constrain output form [6]. RelGate has a narrower aim: it reviews one supplied bundle and requires source text for every passing gate.

II. RELGATE DESIGN

G1 Observability and **G2** Alerting cover detection. **G3** Rollout Safety and **G4** Rollback cover containment and recovery.

This work was conducted independently and does not relate to the author’s employment. No employer resources, data, or confidential information were used.

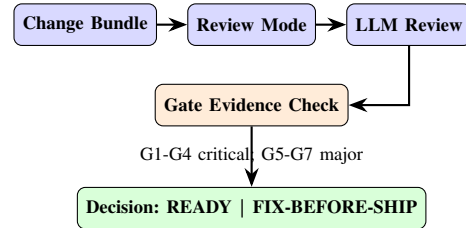


Fig. 1. RelGate workflow. The model must cite exact evidence for each passing gate. Missing critical-gate evidence produces `FIX-BEFORE-SHIP`, while missing major-gate evidence raises a warning.

G5 Ownership covers accountability, **G6** Reliability Impact covers service-level consequences, and **G7** Validation covers pre-release evidence. **G1–G4** block release when evidence is missing. Missing evidence for **G5–G7** raises a warning but does not by itself prevent `READY`. We use these seven gates here. They are not intended as a universal checklist. Organizations can change the gates or their criticality without changing the evidence-citation rule.

For this pilot, a PASS citation had to be a verbatim contiguous span from the supplied bundle after only case and whitespace normalization. The citation could be shorter than its source sentence. Paraphrases, semantic equivalents, and prefix/suffix matches were unsupported. Missing evidence required `MISSING_EVIDENCE`. A matching quote can still be irrelevant or operationally insufficient. We check literal support only.

III. PILOT STUDY

We created 12 scenarios around these gates: 9 with seeded gaps (e.g., absent alerts or rollback) and 3 fully specified `READY` controls. We reviewed each scenario once under **M0 Freeform**, **M1 Checklist** (**G1–G7**), and **M2 RelGate** (gates plus grounding) using GPT-5.5, Grok 4.3, and Llama 4 Maverick via OpenRouter at temperature 0. The experiment comprised $12 \times 3 \times 3 = 108$ calls.

We report Critical Gap Recall, False-Ready Rate (unsafe approvals, $n=27/\text{mode}$), False-Block Rate (`READY` controls blocked, $n=9/\text{mode}$), Decision Accuracy ($n=36/\text{mode}$), and Unsupported Evidence Claims. The Unsupported Evidence Claims metric is the mean of 36 response-level unsupported/total-claim fractions, with a value of zero when

TABLE I
CONTROLLED PILOT RESULTS ACROSS REVIEW MODES.

Metric	M0	M1	M2
Critical Gap Recall	0.975	1.000	1.000
False-Ready Rate ($n=27$)	0.000	0.000	0.000
False-Block Rate ($n=9$)	1.000	0.111	0.000
Unsupported Evid. Claims	0.035	0.208	0.056
Decision Accuracy ($n=36$)	0.750	0.972	1.000

a response makes no claim. Scoring was deterministic. Extracted quotes underwent the strict test above, and PASS on a seeded-missing gate also counted as unsupported. We used no human adjudication. The artifact reports model-level results and Wilson intervals.

IV. PRELIMINARY RESULTS

M0 found nearly all critical gaps (0.975) but blocked every READY control, yielding 0.750 accuracy. M0’s lower unsupported-claim fraction coincided with blocking all READY controls. M1 reduced false blocking to 0.111 and reached 0.972 accuracy, but its unsupported-claim fraction was 0.208. M2 produced no observed false-ready or false-block decisions. Its observed accuracy was 1.000, and its unsupported-claim fraction was 0.056, lower than M1’s.

At model level, each model was 12/12 correct under M2. Its unsupported-claim fractions were 0.138 (Llama), 0.031 (GPT), and 0.000 (Grok), compared with 0.208, 0.083, and 0.333 under M1. In one M1 review of a queue-worker change with no alert evidence, Grok correctly blocked release but presented a paraphrased rollback statement in quotation marks. The decision was correct, but the quoted support was not. We score decision correctness and quotation validity separately.

The sample is small. For M2, 95% Wilson intervals are $[0, 0.125]$ for 0/27 false-ready and $[0, 0.299]$ for 0/9 false-block observations. These are zero observed errors, not zero risk.

V. DISCUSSION AND LIMITATIONS

This small synthetic pilot tests feasibility, not production performance.

The scenarios were built around the same seven gates given to M2. Its 1.000 decision accuracy may therefore reflect compliance with those instructions as much as general readiness reasoning. It should not be read as a production accuracy estimate. Our READY controls were fully specified. Real change records can contain evidence that is ambiguous, stale, conflicting, incomplete, or organization-specific. We observed each case/model/mode once at temperature 0, so repeated-run robustness is unknown. Model versions and hosted implementations may change. Our results apply to the versions and saved outputs evaluated here.

Some gates can be checked partly with deterministic rules, such as whether an alert definition or rollback field is present. RelGate asks whether each passing gate is supported by evidence in a prose/configuration bundle. A deterministic baseline remains future work. A larger study should use real or anonymized change records, repeat each model condition, and include expert/developer review. It should evaluate four aspects separately: decision correctness, quotation validity, semantic relevance, and actual evidence adequacy. It should also test whether evidence is current, mutually consistent, and sufficient under the organization’s own release policy rather than merely present.

Artifact: <https://github.com/lokesh0186/relgate>

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